# DELHIPUBLICSCHOOLJANKIPURAM,LUCKNOW

## SESSION2025-2026:PERIODICASSESSMENT-I

**SET-II**

**Subject:Mathematics**

**TIME : $1\frac{1}{2}$ Hrs.** **M.M. :40**

**Date :**_____ **Name :** _____ **Class/SecX**_____

**General Instructions:**

**Read the following instructions carefully and follow them:**

- **Q 1 to Q10 carry one mark each.**
- **Q11 to Q15are very short type questions and carry two mark seach.**
- **Q16 to Q17 are short type questions and carry three marks each.**
- **Q18 to Q19 are long answer type questions and carry five marks each.**
- **Q20is case based question and is of four marks.**

**********

## SECTION- A

1. The HCF and LCM of two numbers are 7 and 1365 respectively. If one of the number is 91, then what is the difference between the two numbers?

2. Zeroes of a polynomial can be expressed graphically. Number of zeroes of polynomial is equal to number of points where the graph of polynomial is:
   (a) Intersects x-axis (b) Intersects y-axis
   (c) Intersects y-axis or x-axis (d) None of the above

3. If one zero of the quadratic polynomial $x^2 + 3x + k$ is 2, then the value of k is
   (a) 10 (b) –10 (c) 5 (d) –5

4. The Probability of getting a rotten apple in a lot of 400 apples is 0.035, the number of rotten apples in the lot is:
   (a) 07 (b) 14 (c) 21 (d) 28

5. If the probability of an event is t, the probability of its complementary event will be
   (a) t – 1 (b) t (c) 1 – t (d) 1 – 1/t

6. The polynomial equation $x^2 (x + 1) + \sqrt{8} = (x + \sqrt{2})(x^2 - 2)$ is
   (A) linear equation (B) quadratic equation
   (C) cubic equation (D) bi-quadratic equation

**Questions number 7 Assertion and Reason based questions carrying 1 mark. Two statements are given, one labelled as Assertion (A) and the other is labelled as Reason(R). Select the correct answer to these questions from the codes (a), (b), (c) and (d) as givenbelow.**

a) Both assertion and reason are true and reason is the correct explanation of assertion.
b) Both assertion and reason are true but reason is not the correct explanation of assertion.
c) Assertion is true but reason is false.
d) Assertion is false and reason is true.

7. **Assertion:** The graphical representation of a pair of equations $4x + 3y - 1 = 5$ and $12x + 9y = 15$ will be parallel lines.
**Reason:** if the given pair of equations has no solution. That means, the lines representing the given pair of equations are parallel to each other.

c

8. If the discriminant of a quadratic equation is less than zero then it has
a) Equal roots b) real roots c) no real roots d) can't be determined

9. **If one equation of a pair of dependent linear equations is -3x+5y-2=0. The second equation will be:**

(a) -6x+10y-4=0 (b) 6x-10y-4=0
(c) 6x+10y-4=0 (d) -6x+10y+4=0

10. The probability expressed as a percentage of a particular occurrence can never be

(a) less than 100 (b) less than 0
(c) greater than 1 (d) anything but a whole number

## SECTION-B

11. A box consists of 100 shirts of which 88 are good, 8 have minor defects and 4 have major defects. Ramesh, a shopkeeper will buy only those shirts which are good but 'Kewal another shopkeeper will not buy shirts with major defects. A shirt is taken out of the box at random. What is the probability that:
(i) Ramesh will buy the selected shirt?
(ii) 'Kewal will buy the selected shirt?

12. Does $(x - 1)^2 + 2(x + 1) = 0$ have a real root? Justify your answer.

13. Find the zeroes of the following polynomials by factorisation method and verify the relations between the zeroes and the coefficients of the polynomials: $4x^2 - 3x - 1$

14. **Frame equations of the given statement** :Champa went to a 'Sale' to purchase some pants and skirts. When her friends asked her how many of each she had bought, she answered, "The number of skirts is two less than twice the number of pants purchased. Also, the number of skirts is four less than four times the number of pants purchased". Help her friends

Or

Solve the following pair of linear equations: $21x + 47y = 110$; $47x + 21y = 162$

15. As part of their commitment to achieving Sustainable Development Goal 15 (SDG 15) - Life on Land, a community is actively involved in an afforestation project. The main objective of this project is to restore biodiversity and combat land degradation within designated areas, thereby fostering the creation of thriving and balanced ecosystems.

The Eco Warrior Club of Delhi Public School Jankipuram has enthusiastically taken charge of this initiative. They want to plant 66 apple trees, 88 banana trees and 110 mango trees in equal rows (in terms of number of trees). Also, They want to make distinct rows of trees (i.e., only one type of trees in one row). Find the number of minimum rows required

## SECTION-C

16. Prove that $5\sqrt{5} - 3\sqrt{5}$ is an irrational number.

17. A box contains 20 balls bearing numbers 1, 2, 3, 4, 5, ....., 18, 19, 20. A ball is drawn at random from the box. What is the probability that the number on the ball is:

(i) divisible by 2 or 3? (ii) prime number? (iii) not divisible by 10?

## SECTION-D

18. Sita Devi wants to make a rectangular pond on the road side for the purpose of providing drinking water for street animals. The area of the pond will be decreased by 3 square feet if its length is decreased by 2 ft. and breadth is increased by 1 ft. Its area will be increased by 4 square feet if the length is increased by 1 ft. and breadth remains same. Find the dimensions of the pond.

Or

Solve for x : $\frac{1}{(x-1)(x-2)} + \frac{1}{(x-2)(x-3)} = \frac{2}{3}$ , $x \neq 1, 2, 3$

19. Solve the following pair of linear equations graphically:
$x + 3y = 6$ ; $2x - 3y = 12$
Also find the area of the triangle formed by the lines representing the given equations with y-axis.

## SECTION-E

20. The below picture are few natural examples of parabolic shape which is represented by a quadratic polynomial. A parabolic arch is an arch in the shape of a parabola. In structures, their curve represents an efficient method of load, and so can be found in bridges and in architecture in a variety of forms.

a)If the roots of the quadratic polynomial are equal, where the discriminant $D = b^2 - 4ac$, then

i) $D > 0$ ii) $D < 0$ iii) $D \neq 0$ iv) $D = 0$

b) Graph of a quadratic polynomial is a

i) straightline ii) circle iii) parabola (iv)ellipse

c)The representation of Highway Underpass whose one zero is 6 and sum of the zeroes is 0, is

(i)$x^2 - 6x + 2$ (ii) $x^2 - 36$ (iii)$x^2 - 6$ (iv)$x^2 - 3$

(or)

The number of zeroes that polynomial $f(x) = (x-2)^2 + 4$ can have is:

(i)1 ii)2 iii)0 iv) 3